

Sayantan Roy

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Education

Purdue University Ph.D. Candidate in Economics Advisor: Mario J. Crucini	2021–2027 (expected)
Madras School of Economics M.A. in Economics	2017–2019
R.K.M.R. College, Narendrapur B.S. in Economics	2014–2017

Research Fields

Primary: Macroeconomics **Secondary:** International Trade

Job Market Paper

Where Does Fiscal Stimulus Create Jobs? Evidence from U.S. Counties

Abstract. The size of funding and its allocation across states and counties are jointly determined in a national stimulus program. I study whether the second margin of choosing where to spend the money matters for the employment the program creates. Using county-level variation in American Recovery and Reinvestment Act (2009–12) outlays, I find that the local employment responses to government spending are largest in mid-sized counties (approximately 100,000 residents) and substantially smaller in both small counties (approximately 10,000 residents) and large counties (approximately 1 million residents). Evidence on local production structure, commuting and purchasing patterns, and hiring points to two opposing forces. While small counties leak most of the demand they receive to other places, large counties turn the demand they retain into fewer jobs, consistent with greater worker bargaining power and higher recruiting costs. Mid-sized counties balance the two. I interpret and rationalize these patterns using a multi-region New Keynesian model in which locations are linked through trade and workers and firms interact in labor markets characterized by search and matching frictions. I use the model to recover the aggregate employment response (the ‘missing intercept’) that cross-sectional estimates alone cannot identify. In counterfactual reallocations with the aggregate budget unchanged I find that moving 35 percent of large-county spending to mid-sized counties would have created about 145,800 additional job-years over 2009–12 (3.3 percent) and lowered the cost per job-year from \$48,109 to \$46,551. The spatial allocation of stimulus is thus an important dimension of stabilization policy distinct from the size of the program.

Work in Progress

Tariffs Along the Supply Chain: Production Networks and Aggregate Adjustment in the Interwar United States

with Mario J. Crucini

Common Shocks, Local Exposure: Sectoral Wage Dynamics Across U.S. Counties

Gender Gaps in Employment Seasonality: The Role of Unemployment Insurance

Research Experience

Research Assistant , Mario J. Crucini, Purdue University	2023–present
Research Assistant , Seungki Hong, Purdue University	2022

Teaching Experience

Purdue University, Department of Economics

2021–present

Instructor of Record

Summer 2023, 2025

Introductory Macroeconomics (ECON 252)

Teaching Assistant

Macroeconomic Theory III (ECON 612, Ph.D.); Money and Banking (ECON 380);

Introductory Econometrics (ECON 360); Intermediate Macroeconomics (ECON 352)

Industry Experience

American Express

2019–2021

Data Scientist, Credit and Fraud Risk Division

Awards and Fellowships

Krannert Doctoral Students' Research Symposium Winner, Purdue University

2024

Novshek-Watts Scholarship, Purdue University

2021

Graduate Excellence Award, Madras School of Economics

2019

Academic Fellowship, Madras School of Economics

2017–2019

Academic Excellence Award, R.K.M.R. College, Narendrapur

2017

Conference Presentations

Midwest Macroeconomics Meetings, Marquette University

2026

20th Annual Conference on Economic Growth and Development, ISI Delhi

2025

Computational Skills

Python, JAX, Julia, MATLAB, Stata, SQL, PySpark, Git

References

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Cathy M. Zhang

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